PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Safety and Efficacy of Acupuncture for Varicocele-Induced Male Infertility: A systematic review protocol
AUTHORS	wang, sijia; Lu, Rongchen; Shi, Hongshuo; Chen, Jiangnan; Sun, Miaomiao; Ding, Jing; Lv, Qiang; Wang, Chenyao; Ren, Jianjun; Zhou, Guangming; Tang, Zhian

VERSION 1 – REVIEW

REVIEWER	Jesse Mills David Geffen School of Medicine, Urology
REVIEW RETURNED	26-Apr-2022

GENERAL COMMENTS	The authors set out a plan to perform a systematic review evaluating the efficacy of the treatment of varicocele for infertility with acupuncture. From a content standpoint, the authors have posed a study to accurately evaluate the current literature on this topic. Paper would benefit from minor English grammatical review as there are some nuances in wording that should be corrected for better flow.

REVIEWER	Jiaman Wu Shenzhen Maternity & Child Healthcare Hospital
REVIEW RETURNED	28-Apr-2022

GENERAL COMMENTS	1. This study has not registered, a registration is needed. 2. Because the article mentioned that the anatomical basis of the disease needs to be fundamentally solved in order to make the results satisfactory, it is necessary to explain why acupuncture and moxibustion treatment of VC is selected as a scientific problem. 3.The statistical results is too superficial, mainly due to the indistinct purpose of this study. The extracted outcomes are insufficient and the utilization of data is low. For example, subgroup analyses could be conducted by different type of acupuncture methods or different follow-up time. Furthermore, sensitivity analyses can be performed. 4.There are many problems in the paper, such as unclear logic, vague concept, lack of standardization and rigor. The search string is too simple, is there any supplementary attachment? The description of inclusion and exclusion criterion are not clear, the included control group and experimental group should be clearly listed. The screening process does not list the screening results of each procedure (primary and full-text screening). The allocation and blind methods of some studies are unclear, it's better to contact the author to confirm if allowed. 5.Due to the lack of adequate statistical analyses, the conclusions in the discussion are far from profound.

REVIEWER	Robin Prescott University of Edinburgh, Medical Statistics Unit
REVIEW RETURNED	27-Jun-2022

GENERAL COMMENTS	I am reviewing this protocol as a statistician though I will also make comments of a more general nature about other details of the protocol. Because of my main focus I will concentrate initially on the data collection and analysis part of the protocol. The general approach for reviewing the articles with two researchers undertaking each stage, with a sensible method for resolving inconsistencies, is entirely sensible. Contacting the corresponding authors with any queries arising from the paper is always sensible. I would take that a stage further than the authors indicate and would attempt to contact them if there was any aspect of the presentation that was not clear to me, rather than only seeking missing details. Using the Cochrane risk of bias tool is a good choice for evaluating bias. It is worth noting that within recent months the Cochrane Editorial Board has stated that it now expects Cochrane reviews to be implementing Risk of Bias 2. Having assessed the risk of bias in contributing studies, the authors have not yet mentioned how that assessment would be used in performing meta analyses. Is there a cut-off at which the authors would determine that a study has too great a risk of bias for inclusion in a meta analysis? Might they perform sensitivity analyses with exclusion of the studies at high risk of bias? I think that the protocol should be indicating how the risk of bias will affect the subsequent analyses and interpretation. The current protocol suggests the use of fixed effects or random effects model for the meta analyses, dependent on the value of I-squared. In contrast, I would advocate the use of random effects models for all meta analyses of this kind. They will be dependent upon individual randomised controlled trials conducted in a variety of locations with separate protocols that will not be identical, and with different populations of patients. Under such circumstances I think it is an unreasonable assumption that the 'true' effect sizes from every contributing study will be identical, even if it is found in practice that the value of I-squared is less than a pre-specified cut-off. I see no justification for the use of fixed effects models in any situations that the authors of the protocol are likely to meet. The sentence on unit of analysis issues is not really saying anything useful. I only see this being an issue if a suitable cluster randomised trial is identified. If such a trial is reported with standard errors that reflect the clustering of observations, there is no substantial problem. If that is not the case, I suggest that the authors follow the guidance in Higgins JPT, Eldridge S, Li T (editors). Chapter 23: Including variants on randomized trials. In: Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). Cochrane Handbook for Systematic Reviews of Interventions version 6.3 (updated February 2022). Cochrane, 2022. Apart from those issues already identified, the rest of the section on data collection and analysis is reasonable. Turning to design issues, I understand that varicocele can occur without causing infertility. It would therefore be helpful for the authors to explain why their study is restricted to trials recruiting

	patients with varicocele infertility. On a similar theme, inclusion criterion 2, that the patient had unprotected, regular intercourse for at least one year with a healthy partner is sensible for the secondary outcome of pregnancy rates, but it not evident that it is necessary for the primary outcome of results from a routine semen analysis. In practice, trials may occur where the majority of patients satisfy the authors' inclusion and exclusion criteria, but there will be occasional departures from these criteria. For example, perhaps the age range is 21-50, or a small number of fertile patients were included, or an occasional patient has bilateral varicocele. Will a single failure of compliance to the inclusion/exclusion criteria result in exclusion of a study, or will some tolerance be allowed? It is a dilemma that should be addressed now and clarified in the protocol. I am not an information specialist, but I am aware of some of the recommendations that appear in the Cochrane Handbook in relation to searching for and selecting studies. In particular, I think it is important that those designing systematic reviews consult an information specialist or a medical/healthcare librarian with experience in supporting systematic reviews. When reviewing systematic reviews submitted to journals, that is a recommendation I make frequently, and it commonly results in resubmissions containing a more comprehensive coverage of the literature. The authors mention searching the Cochrane Library. I think they will find it particularly helpful to search CENTRAL (The Cochrane Central Register of Controlled Trials). For additional details I highly recommend that the authors consult Lefebvre C, Glanville J, Briscoe S, Featherstone R, Littlewood A, Marshall C, Metzendorf M-I, Noel-Storr A, Paynter R, Rader T, Thomas J, Wieland LS. Chapter 4: Searching for and selecting studies. In: Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). Cochrane Handbook for Systematic Reviews of Interventions version 6.3 (updated February 2022). Cochrane, 2022. Available from www.training.cochrane.org/handbook. There is a minor issue. The Methods and analysis section has a February 2022 cut-off for publication of relevant articles. That should be amended to a more current date. Finally, the quality of the English is capable of improvement. The intention of the authors is generally clear, but there is a sentence in the second sentence of the Abstract when the intent of the authors is hard to determine. The sentence ends "but the benefits of surgical repair can be very slightly". Please explore with an expert in the English language, precisely what is meant here.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1:
Dr. Jesse Mills, David Geffen School of Medicine
1. Comment: Paper would benefit from minor English grammatical review as there are some nuances in wording that should be corrected for better flow.
Response: We are very sorry for our negligence of problems with English grammar, nuances in wording and language fluency in texts. We had the text proofread by a language expert from a translation company to ensure that the paper was largely free of typos, grammatical and syntactical errors, and poor word choice. Certificate of English Proofreading is attached to the supplementary document.

Reviewer: 2
Dr. Jiaman Wu, Shenzhen Maternity & Child Healthcare Hospital
1. Comment: 1. This study has not registered, a registration is needed.
Response: It is really true as Reviewer suggested that a registration is needed. We have registered and updated our registration number. (Line 68, page 4).

2. Comment: 2. Because the article mentioned that the anatomical basis of the disease needs to be fundamentally solved in order to make the results satisfactory, it is necessary to explain why acupuncture and moxibustion treatment of VC is selected as a scientific problem.
Response: We are grateful for the suggestion. To be clearer and in accordance with the reviewer concerns, we have added a brief description. (Line 56-59, page 3).

3.Comment: 3. The statistical results is too superficial, mainly due to the indistinct purpose of this study. The extracted outcomes are insufficient and the utilization of data is low. For example, subgroup analyses could be conducted by different type of acupuncture methods or different follow-up time. Furthermore, sensitivity analyses can be performed.
Response: Thank you for pointing this out. It is true that previously we did not extract the results sufficiently and make the most of the extracted data. We have revised and adjusted the content of the outcome indicators and outcome measures. (Line 99-106, page 5). We also discuss this part in more detail and added the suggested content to the manuscript. (Line 173-174, page 10).

4. Comment: 4. There are many problems in the paper, such as unclear logic, vague concept, lack of standardization and rigor. The search string is too simple, is there any supplementary attachment? The description of inclusion and exclusion criterion are not clear, the included control group and experimental group should be clearly listed. The screening process does not list the screening results of each procedure (primary and full-text screening). The allocation and blind methods of some studies are unclear, it's better to contact the author to confirm if allowed.
Response: We are very sorry for these problems pointed out by the reviewer. We should think more thoroughly. We modified the search strategy according to the Cochrane Handbook and have an experienced reviewer check out our search strategy. Full search strategies for all databases, registers and websites are shown in supplementary file. (Line 111-118, page 6). Details of the included control group and experimental group are listed in "Types of interventions" part. (Line 91-95, page 5) Concerning the screening results of each procedure (primary and full text screening), we will present these in a subsequent systematic review if possible, and it is acceptable not to present the specific screening results in this protocol. Corresponding authors will be contacted to obtain the missing details in the allocation and blinded sections. (Line 130-132, page 8).

5. Comment: 5. Due to the lack of adequate statistical analyses, the conclusions in the discussion are far from profound.
Response: Thank you for the above comments and suggestions. The reviewer's comments were very pertinent and helpful. If possible, we will present statistical analyses results and conclusions in a subsequent systematic review; it is acceptable not to present specific analyses results and conclusions in this protocol.

Reviewer: 3
Prof. Robin Prescott, University of Edinburgh
1. Comment: I am reviewing this protocol as a statistician though I will also make comments of a more general nature about other details of the protocol.
Because of my main focus I will concentrate initially on the data collection and analysis part of the protocol. The general approach for reviewing the articles with two researchers undertaking each stage, with a sensible method for resolving inconsistencies, is entirely sensible. Contacting the

corresponding authors with any queries arising from the paper is always sensible. I would take that a stage further than the authors indicate and would attempt to contact them if there was any aspect of the presentation that was not clear to me, rather than only seeking missing details.
Response: We gratefully thanks for the precious time the reviewer spent making valuable remarks. We consider this suggestion to be of great help to our future practice and we have revised this the data collection and analysis part according to the Reviewer's suggestion. (Line 131-132, page 8).

2. Comment: Using the Cochrane risk of bias tool is a good choice for evaluating bias. It is worth noting that within recent months the Cochrane Editorial Board has stated that it now expects Cochrane reviews to be implementing Risk of Bias 2. Having assessed the risk of bias in contributing studies, the authors have not yet mentioned how that assessment would be used in performing meta analyses. Is there a cut-off at which the authors would determine that a study has too great a risk of bias for inclusion in a meta analysis? Might they perform sensitivity analyses with exclusion of the studies at high risk of bias? I think that the protocol should be indicating how the risk of bias will affect the subsequent analyses and interpretation.
Response: We are very sorry for our negligence of applying the updated Cochrane risk of bias tool. We have rewritten this "Assessment of risk of bias" section according to the reviewer's comments. If a RCT was judged to have "some concerns" about the risk of bias in three or more domains, it will be excluded from the systematic review due to the high risk of bias. (Line 136-141, page 8-9).

3. Comment: The current protocol suggests the use of fixed effects or random effects model for the meta analyses, dependent on the value of I-squared. In contrast, I would advocate the use of random effects models for all meta analyses of this kind. They will be dependent upon individual randomised controlled trials conducted in a variety of locations with separate protocols that will not be identical, and with different populations of patients. Under such circumstances I think it is an unreasonable assumption that the 'true' effect sizes from every contributing study will be identical, even if it is found in practice that the value of I-squared is less than a pre-specified cut-off. I see no justification for the use of fixed effects models in any situations that the authors of the protocol are likely to meet.
Response: Thank you for pointing out this problem in manuscript and thank you for your nice suggestion. We were not aware enough of this issue before and we have rewritten this section in "Subgroup analysis and investigation of heterogeneity" according to the reviewer's comments. We will use the random effects model for the meta analysis. (Line 168-174, page 10).

4. Comment: The sentence on unit of analysis issues is not really saying anything useful. I only see this being an issue if a suitable cluster randomised trial is identified. If such a trial is reported with standard errors that reflect the clustering of observations, there is no substantial problem. If that is not the case, I suggest that the authors follow the guidance in Higgins JPT, Eldridge S, Li T (editors). Chapter 23: Including variants on randomized trials. In: Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). Cochrane Handbook for Systematic Reviews of Interventions version 6.3 (updated February 2022). Cochrane, 2022.
Response: We are very grateful to the reviewer for the constructive comments. The reviewer's comment has inspired our thoughts. We followed the guidance in Higgins JPT, Eldridge S, Li T (editors). Chapter 23: Including variants on randomized trials and we have re-written this part "The issue concerning the unit of analysis". (Line 148-152, page 9).

5. Comment: Turning to design issues, I understand that varicocele can occur without causing infertility. It would therefore be helpful for the authors to explain why their study is restricted to trials recruiting patients with varicocele infertility. On a similar theme, inclusion criterion 2, that the patient had unprotected, regular intercourse for at least one year with a healthy partner is sensible for the secondary outcome of pregnancy rates, but it not evident that it is necessary for the primary outcome of results from a routine semen analysis. In practice, trials may occur where the majority of patients satisfy the authors' inclusion and exclusion criteria, but there will be occasional departures from these

criteria. For example, perhaps the age range is 21-50, or a small number of fertile patients were included, or an occasional patient has bilateral varicocele. Will a single failure of compliance to the inclusion/exclusion criteria result in exclusion of a study, or will some tolerance be allowed? It is a dilemma that should be addressed now and clarified in the protocol.
Response: We totally understand the reviewer's concern. It is true that we do need to explain why our study is restricted to trials recruiting patients with varicocele infertility. We have inserted this section in the discussion section. (Line 197-200, page 11). Inclusion criterion 2 was to ensure that patients were infertile and, in addition, that regular seminal discharge had an effect on the results of routine semen test. Regarding the age of patients included, we have made appropriate modifications in the light of the reviewer's comments. (Line 76-79, page 4). Because of the similarity of the pathogenesis, we will not exclude RCTs that include the occasional patients with bilateral varicoceles. (Line 76,81, page 4). In order to ensure practicability, we have allowed some tolerance in the inclusion and exclusion criteria. We add the content in Line 79-80, page 4.

6. Comment: I am not an information specialist, but I am aware of some of the recommendations that appear in the Cochrane Handbook in relation to searching for and selecting studies. In particular, I think it is important that those designing systematic reviews consult an information specialist or a medical/healthcare librarian with experience in supporting systematic reviews. When reviewing systematic reviews submitted to journals, that is a recommendation I make frequently, and it commonly results in resubmissions containing a more comprehensive coverage of the literature. The authors mention searching the Cochrane Library. I think they will find it particularly helpful to search CENTRAL (The Cochrane Central Register of Controlled Trials). For additional details I highly recommend that the authors consult Lefebvre C, Glanville J, Briscoe S, Featherstone R, Littlewood A, Marshall C, Metzendorf M-I, Noel-Storr A, Paynter R, Rader T, Thomas J, Wieland LS. Chapter 4: Searching for and selecting studies. In: Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). Cochrane Handbook for Systematic Reviews of Interventions version 6.3 (updated February 2022). Cochrane, 2022. Available from www.training.cochrane.org/handbook.
Response: We found the reviewer's suggestions very useful. The coverage of the previous search strategy was not comprehensive. As Reviewer suggested that we modified the search strategy according to the Cochrane Handbook and have an experienced reviewer check out our search strategy. We have changed our search from the Cochrane Library to CENTRAL (The Cochrane Central Register of Controlled Trials). The selection of English databases has also been adjusted according to Chapter 4: Searching for and selecting studies. (Line 109-118, page 7). Full search strategies for all databases, registers and websites are shown in supplementary file.

7. Comment: There is a minor issue. The Methods and analysis section has a February 2022 cut-off for publication of relevant articles. That should be amended to a more current date.
Response: Thank you so much for your careful check. We have amended it to a more current date. (Line 109, page 6).

8. Comment: Finally, the quality of the English is capable of improvement. The intention of the authors is generally clear, but there is a sentence in the second sentence of the Abstract when the intent of the authors is hard to determine. The sentence ends “but the benefits of surgical repair can be very slightly”. Please explore with an expert in the English language, precisely what is meant here.
Response: We are very sorry for our incorrect writing and we have amended the second sentence of the Abstract. (Line 13-14, page 1). We had the text proofread by a language expert from a translation company to ensure that the paper was largely free of typos, grammatical and syntactical errors, and poor word choice. Certificate of English Proofreading is attached to the supplementary document.

VERSION 2 – REVIEW

REVIEWER	Robin Prescott University of Edinburgh, Medical Statistics Unit
REVIEW RETURNED	17-Aug-2022

GENERAL COMMENTS	I thank the authors for their changes to the protocol. There is a very considerable improvement, and I now have only some points of detail to bring to the attention of the authors. 1. In the final paragraph of the Introduction there is repetition when the lack of a systematic review is mentioned twice in the same sentence. 2. In the third line of page 7, PROSPERO is spelt incorrectly. The same spelling error occurs again at the very end of the Supplementary material. 3. In the section on Data collection and analysis, there is inconsistency in the tenses used. Rather than “imported” I suggest “will import”. Replace “met” with “meet”. 4. In the Assessment of risk of bias section, in the final sentence, replace ”it was excluded” with “it will be excluded”. 5. In the section on Summary of evidence, the first sentence ends by stating that with less than 10 studies, the reporting bias detection is not necessary. It is not so much that it is unnecessary, but that detection of reporting bias is infeasible. I suggest that the section after the semi colon is replaced by “if not, the detection of the reporting bias is infeasible”. Apart from these very minor issues, I have no further comments.

VERSION 2 – AUTHOR RESPONSE

Reviewer: 3

Prof. Robin Prescott, University of Edinburgh

Comments to the Author:

1. In the final paragraph of the Introduction there is repetition when the lack of a systematic review is mentioned twice in the same sentence.

Response: We are very grateful for your careful checking. We have removed the repetitive expression. (Page 3, Line 62)

2. In the third line of page 7, PROSPERO is spelt incorrectly. The same spelling error occurs again at the very end of the Supplementary material.

Response: We are sorry for the mistakes we have made and we have corrected this spelling error in the main manuscript (Page 8, Line 120) and in the Supplementary material.

3. In the section on Data collection and analysis, there is inconsistency in the tenses used. Rather than “imported” I suggest “will import”. Replace “met” with “meet”.

Response: Thank you so much for your careful check. We have corrected the tense issues. (Page 8, Line 125,126)

4. In the Assessment of risk of bias section, in the final sentence, replace “it was excluded” with “it will be excluded”.

Response: We feel sorry for the having ignored the tense issue. We have corrected this mistake. (Page 8, Line 141)

5. In the section on Summary of evidence, the first sentence ends by stating that with less than 10 studies, the reporting bias detection is not necessary. It is not so much that it is unnecessary, but that detection of reporting bias is infeasible. I suggest that the section after the semi colon is replaced by “if not, the detection of the reporting bias is infeasible”.

Response: We gratefully appreciate for your valuable suggestion. We have amended this sentence as you suggested. (Page 10, Line 183)